R-INVARIANT AFFINE THICKENING PAIRS AROUND GENERIC ORBIT POINTS FOR QUASI-SEPARATED U WITH FINITE FINITELY PRESENTED SOURCE AND TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The R-invariant thickening pair W' inside W with finite locally free restriction and the covering property (more-groupoids chapter, Lemma lemma-finite-flat-over-almost-dense-subscheme, canonical tag 0ABA; collected on this lane as L22511). cited at proof lines 2816-2817. Provides the pair W' inside W driving the reduction.
- Addendum: with s, t of finite presentation the thickening W' to W is of finite presentation and generic-orbit points lie in W (more-groupoids chapter, Lemma lemma-finite-flat-over-almost-dense-subscheme-addendum, canonical tag 0ABB; collected on this lane as L22521). cited at proof lines 2818-2819. Gives u_j in W and the finite presentation of the thickening, the two inputs making the reduction work.
- Criterion for gluing affines inside a closed affine (limits chapter, Lemma limits-lemma-affines-glued-in-closed-affine, canonical tag 09NL). cited at proof lines 2820-2823. Reduces the task to finding an R-invariant affine open V' of W'; the corresponding V inside W is then affine.
- Affine opens containing finitely many maximal-type points of a quasi-separated scheme (properties chapter, Lemma properties-lemma-maximal-points-affine, canonical tag 01ZV). cited at proof lines 2828-2830. Finds an affine open containing the union of the orbits in the restricted situation.
- Existence of R-invariant affine opens containing finitely many points for groupoids with finite locally free structure maps (groupoids chapter, Lemma groupoids-lemma-find-invariant-affine, canonical tag 03JE). cited at proof lines 2830-2832. Produces the final R-invariant affine open containing the orbit points.

Source reference. Original lemma 0ABA: sources/stacks-chow/tags.txt, source lines 13013–13013. Cited at proof lines 2816-2817: supplies the R-invariant pair W' inside W.

Date: stacks-project:more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0ABB: sources/stacks-chow/tags.txt, source lines 13014–13014. Cited at proof lines 2818-2819: u_j in W and the thickening of finite presentation.

Source reference. Original lemma 09NL: sources/stacks-chow/tags.txt, source lines 12219–12219. Cited at proof lines 2820-2823: reduces the construction to an R-invariant affine open of W'.

Source reference. Original lemma 01ZV: sources/stacks-chow/tags.txt, source lines 2610–2610. Cited at proof lines 2828-2830: affine open containing the union of the orbits.

Source reference. Original lemma 03JE: sources/stacks-chow/tags.txt, source lines 4636–4636. Cited at proof lines 2830-2832: produces the R-invariant affine open for the finite locally free restriction.

Lemma 0.2. *Let (U, R, s, t, c) be a groupoid scheme over a scheme S. Assume s, t are finite and of finite presentation and U quasi-separated. Let $u_1, \ldots, u_m \in U$ be points whose orbits consist of generic points of irreducible components of U. Then there exist R-invariant subschemes $V' \subset V \subset U$ such that*

(1) *$u_1, \ldots, u_m \in V'$,*
(2) *V is open in U,*
(3) *V' and V are affine,*
(4) *$V' \subset V$ is a thickening of finite presentation,*
(5) *the morphisms s', t' of the restriction (V', R', s', t', c') are finite locally free.*

Proof. Let $W' \subset W \subset U$ be as in Lemma 0ABA. By Lemma 0ABB we get $u_j \in W$ and that $W' \to W$ is a thickening of finite presentation. By Limits, Lemma 09NL it suffices to find an R-invariant affine open subscheme V' of W' containing u_j (because then we can let $V \subset W$ be the corresponding open subscheme which will be affine). Thus we may replace (U, R, s, t, c) by the restriction (W', R', s', t', c') to W'. In other words, we may assume we have a groupoid scheme (U, R, s, t, c) whose morphisms s and t are finite locally free. By Properties, Lemma 01ZV we can find an affine open containing the union of the orbits of $u_1, \ldots, u_m$. Finally, we can apply Groupoids, Lemma 03JE to conclude. □